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Deadline: Noon Mar 9, 2026.

1. **[10 points]** *Union-Find models dynamic connectivity, which arises in network routing, image segmentation, and Kruskal's MST algorithm.*

- **Union by rank:** always attach the root of the lower-rank tree under the root of the higher-rank tree; if ranks are equal, choose either root and increment its rank by 1.
- **Path compression:** during $\text{Find}(x)$, make every node on the path from x to the root point directly to the root.

Perform the following operations in order using union by rank. After *each* Union, draw the resulting forest. For every node shown, label its **rank** and indicate the **parent pointer** (or mark it as a root).

 $\text{Union}(1, 3), \quad \text{Union}(5, 7)$

Then perform **Find**(8) without path compression. Write down the sequence of nodes visited while traversing parent pointers from 8 to the root. Draw the resulting forest.

Implement a **FIFO queue** using two stacks S_{in} and S_{out} , each supporting **Push** and **Pop** in $O(1)$ worst case, as follows:

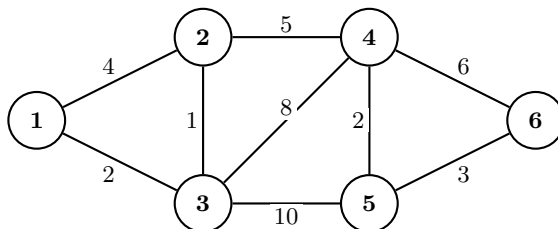
- (a) [**10 pts**] Trace the data structure on the following sequence of operations. After each operation write the contents of S_{in} and S_{out} (top element listed first) and the value returned (if any).

Operation	S_{in} (top first)	S_{out} (top first)	Returned
Enqueue(1)			—
Enqueue(2)			—
Enqueue(3)			—
Dequeue			
Enqueue(4)			—
Dequeue			
Dequeue			

¹Hint: Charge **3 credits** per **Enqueue** and **1 credit** per **Dequeue**, and maintain the **credit invariant**: every element currently in S_{in} holds exactly 2 banked credits.

3. [20 points] *Running Kruskal's algorithm by hand builds intuition for the cut property and reveals how sensitive the MST is to edge weight changes.*

Consider the following weighted undirected graph G with vertices $\{1, 2, 3, 4, 5, 6\}$ and edges as shown.



- (a) [10 pts] List all 9 edges in non-decreasing order of weight (break ties by smaller endpoint first), then trace Kruskal's algorithm. Fill in every row of the table. In the "Action" column write **Add** or **Skip**. In the "Reason" column, for a **Skip** name the cycle that would have been created; for an **Add** state which two components merge.

Step	Edge	Weight	Action	Reason / Components after
1				
2				
3				
4				
5				
6				
7				
8				
9				

- (b) [10 pts] When Kruskal's adds edge $(2, 4)$, it merges two components S and $V \setminus S$. Identify the cut $(S, V \setminus S)$ at that moment, list all edges crossing it with their weights, and verify using the **cut property** that $(2, 4)$ must belong to every MST of G .

Then suppose the weight of $(2, 4)$ is changed to w' . Find the exact threshold below which the MST is unchanged. For $w' = 9$, identify which edge replaces $(2, 4)$, draw the new MST, and state its total weight.